



Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY ADVANCED BOOTCAMP · 40 HOURS

Azure AI Foundry & Semantic Kernel

Master Microsoft's new unified AI platform with deep orchestration

5

DAYS

40

HOURS

25+

HANDS-ON LABS

8

AI PROJECTS

Course Overview

This 5-day advanced bootcamp covers the complete Azure AI Foundry platform — Microsoft's new unified AI development environment that replaces Azure OpenAI Studio. Deep-dive into Semantic Kernel for AI orchestration, build production-grade agents, implement advanced RAG patterns, and master the entire AI application lifecycle from model catalog to deployment.

The most comprehensive Azure AI Foundry training available. Be among the first to master Microsoft's revolutionary new platform. This bootcamp goes beyond simple API calls to teach complete AI system architecture using Semantic Kernel as the orchestration layer. You'll build 8 production-ready solutions leveraging model catalog, prompt flow, evaluation systems, and managed endpoints. Delivered by a Microsoft Certified Trainer with 16+ active Microsoft certifications.

Who Is This For?



AI Engineers

Building production AI systems on Azure



Solution Architects

Designing enterprise AI platforms



DevOps Engineers

Deploying AI infrastructure



Technical Leads

Leading AI transformation initiatives

Learning Outcomes

- 1 Navigate Azure AI Foundry** — Master the unified workspace, understand AI hubs vs projects, leverage model catalog with 1,500+ models
- 2 Build with Semantic Kernel** — Create sophisticated AI applications using plugins, planners, native functions, and memory stores
- 3 Implement Enterprise RAG** — Design production retrieval systems with vector search, hybrid ranking, and evaluation metrics
- 4 Deploy Production Systems** — Implement managed endpoints, content filtering, monitoring, and cost optimization

Prerequisites

Programming experience (Python or C#), basic understanding of AI/ML concepts, Azure fundamentals. Experience with REST APIs and cloud architecture helpful.

5-Day Curriculum

Hands-on format with live coding, practical labs, and real-world projects. Schedule and session order are fully customizable based on your organization's needs.

01

Azure AI Foundry Fundamentals

PLATFORM ARCHITECTURE & SETUP

SESSION A

Platform Introduction

- Azure AI Foundry vs Azure OpenAI Studio migration
- Understanding AI Hubs and Projects architecture
- Model Catalog: exploring 1,500+ models
- Workspace setup and configuration
- Resource provisioning and management

SESSION B

Core Components

- Model deployments and endpoints
- Prompt flow designer walkthrough
- Data connections and datastores
- Compute resources configuration
- Lab: First AI Foundry project deployment

02

Semantic Kernel Deep-Dive

ORCHESTRATION FRAMEWORK MASTERY

SESSION A

SK Fundamentals

- Semantic Kernel architecture overview
- Kernel initialization and configuration
- Semantic functions vs Native functions
- Prompt templates and variables
- Context and memory management

SESSION B

Advanced Orchestration

- Building custom plugins
- Planner implementation (Sequential/Stepwise)
- Function chaining and composition
- Skill orchestration patterns
- Lab: Multi-step reasoning agent

03

Advanced RAG & Vector Systems

PRODUCTION RETRIEVAL ARCHITECTURES

SESSION A**RAG Architecture**

- Document processing pipelines
- Chunking strategies and optimization
- Embedding models selection
- Vector database integration (AI Search)
- Hybrid search implementation

SESSION B**Advanced Patterns**

- Multi-modal RAG systems
- Semantic ranking and reranking
- Query expansion techniques
- Citation and grounding
- Lab: Enterprise knowledge base

04

Prompt Flow & Agent Development

VISUAL DEVELOPMENT & COMPLEX WORKFLOWS

SESSION A**Prompt Flow Designer**

- Visual flow development
- Node types and connections
- Conditional logic and branching
- Tool integration and custom tools
- Testing and debugging flows

SESSION B**Multi-Agent Systems**

- Agent communication protocols
- Supervisor agent patterns
- Tool-use and function calling
- State management in conversations
- Lab: Customer service agent system

05

Production Deployment & Operations

ENTERPRISE-GRADE AI SYSTEMS

SESSION A**Deployment Strategies**

- Managed online endpoints
- Blue-green deployments
- A/B testing for models
- Load balancing and scaling
- Cost optimization strategies

SESSION B**Operations & Monitoring**

- Content safety implementation
- Responsible AI dashboard
- Performance monitoring
- Model evaluation metrics
- Final project presentation

Flexible & Customizable: Session order, timing, depth, and focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest Azure AI Foundry features.

Ready to Master Azure AI Foundry?

Join this intensive 5-day bootcamp to become an expert in
Microsoft's latest AI platform and Semantic Kernel orchestration.

 www.gennoor.com



 +91 9326 352 241

 linkedin.com/company/gennoor-tech-private-limited

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